

Weiyou Li

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 Weiyou Li

 BMath, Pure Mathematics, Statistics, [University of Waterloo](#), Expected 2027

Summary of Qualifications

- **Machine Learning & AI:** Experience applying ML to medical imaging, predictive modeling, and LLM-based applications, including RAG and sampling techniques.
- **Statistical & Quantitative Modeling:** Strong foundation in regression, time-series analysis, stochastic modeling, survival analysis, and experimental evaluation.
- **Data Engineering & Tools:** Experience with Python, SQL, Git, ML experimentation, data pipelines, and model evaluation using PyTorch, TensorFlow, Scikit-learn, lifelines, and Power BI.

Publications & Preprints

Counterexamples Regarding Elementary Symmetric Partitions

2026

Weiyou Li, et al. — Preprint: [arXiv:2606.00420](#) [[math.CO](#)]

Work Experience

Research Assistant

Toronto, ON

Unity Health Toronto — TUPIL, PI: [Dr. Eno Hysi](#)

Jan 2026 – Present

- Developed and evaluated **ResNet-based CNN models** for kidney fibrosis classification by applying transfer learning, cross-validation, and multiple ultrasound input configurations across **3,200+ images**, enabling systematic comparison of learned image features against clinical benchmarks for medical AI research.
- Modeled longitudinal **eGFR trajectories** for **150+ patients** by cleaning raw clinical Excel data, aligning patient timelines, and applying rule-based filtering, **GAMM**, and polynomial fitting, extracting baseline kidney-function and slope-based features to support downstream kidney-function prediction.
- Built automated **Python data-ingestion and quality-control pipelines** to synchronize ultrasound data across cloud and local storage, standardize file naming, validate imaging inventories, and detect missing or duplicate segmentation files, reducing manual ingestion time by approximately **70%** across **730+ patients**.

Statistics Tutor

Etobicoke, ON

Humber Polytechnic

May 2025 – Aug 2025, Sept 2024 – Dec 2024

- Applied **statistical modeling, time-series analysis**, and debugging support to improve student learning, increasing engagement by 17%, improving performance by 15%, and achieving a **4.9/5 satisfaction** rating across 100+ sessions.

Research & Technical Projects

Graduate Admission Prediction (*PyTorch, TensorFlow, Scikit-learn*)

Graduate-Admission

Supervised by Yuan Wu (Duke University)

June 2025 - Sept 2025

- Developed and compared **Artificial Neural Network (ANN)** and **Logistic Regression** models on a 400 sample dataset to **optimize algorithm selection** for low-data scenarios.
- Achieved a 6% increase in AUC and a 10% increase in accuracy by conducting a **hyperparameter optimization** study focused on hidden layer dimensionality, epoch count, and batch sizing.
- Applied **Pearson correlation analysis** to identify and utilize the top 5 predictive features, significantly **enhancing the interpretability** of the final Logistic Regression model.

Beyond VaR: ETF Drawdown Recovery & Survival Risk Modeling

Beyond VaR

(*Python, Pandas, scikit-learn, lifelines, Matplotlib, Financial Backtesting*)

April 2026 - May 2026

- Built an end-to-end **quantitative risk framework** across multiple ETFs to model drawdowns, VaR breaches, and **time-to-recovery risk** beyond traditional VaR to support portfolio allocation decisions and demonstrated ML performance in low-data environments.
- Engineered return, volatility, drawdown, VaR-breach, and recovery-event features, applied **Kaplan–Meier curves, Cox models, and group comparisons** across risk regimes.
- Combined survival-risk scores with statistical and machine learning return predictions to compare **baseline, risk-filtered, and volatility-targeted long/cash strategies**.
- Evaluated performance against buy-and-hold using **Sharpe, Sortino, maximum drawdown, VaR, Expected Short-fall, and recovery-time metrics**.